

Robust method for detecting convergent shifts in evolutionary rates

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Supplementary Figures

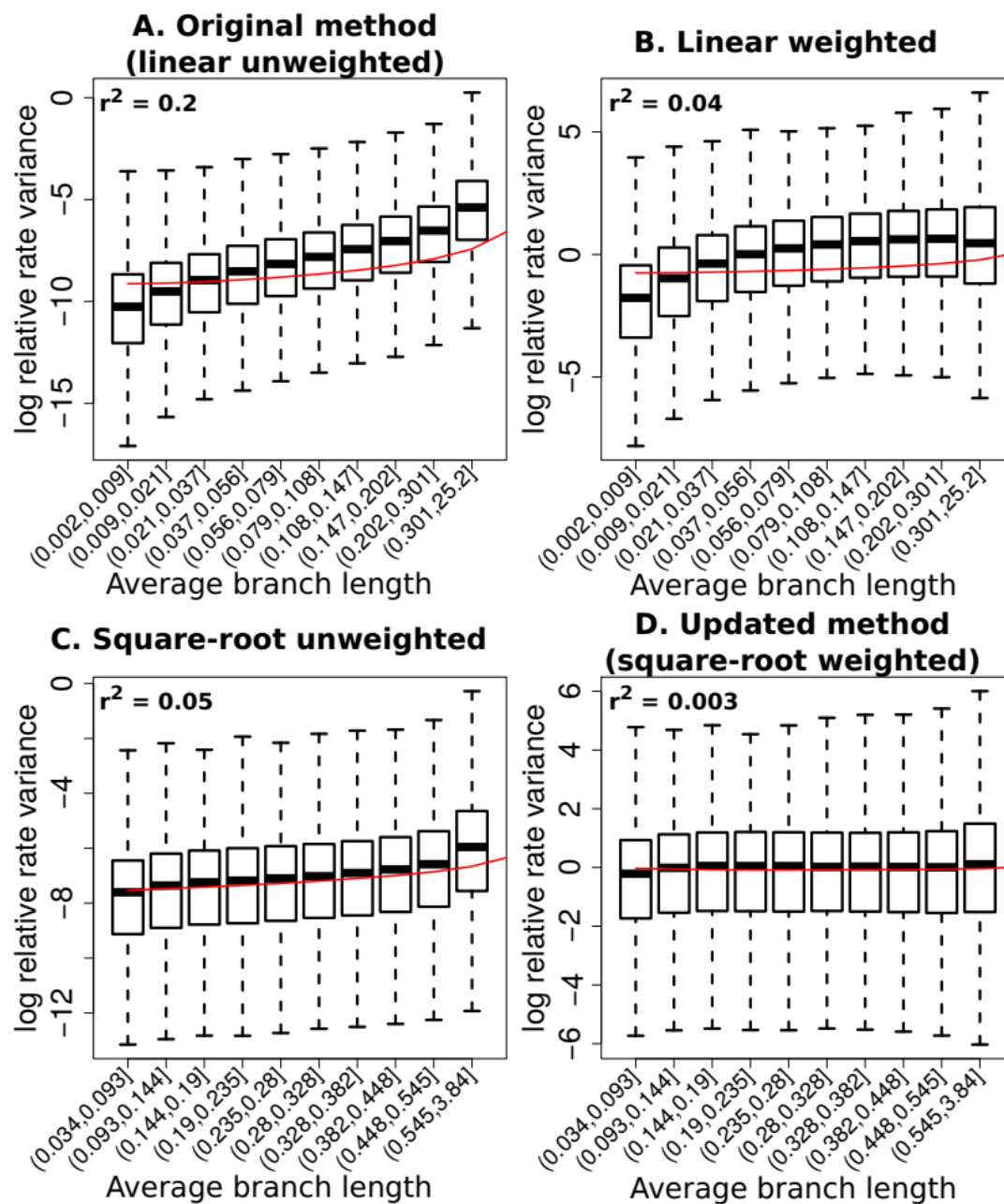


Fig S1. Comparison of mean-variance trends in relative rates computed using original, updated and intermediate methods. A corresponds to original method, D the updated method. Panels B and C reflect methods that are intermediate to the updated method, with no transformation (B), and no weighted regression (C).

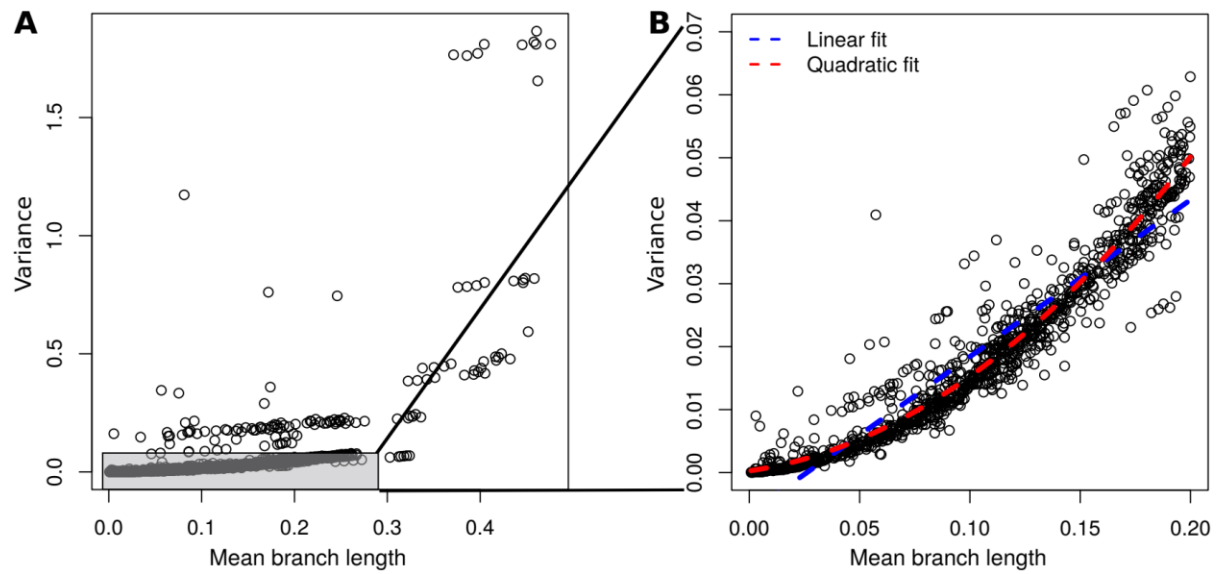


Fig S2. Mean-Variance trends in branch lengths of 19,149 protein-coding gene trees across 63 mammalian species. Panel A represents the full data, and B shows a zoomed in version containing 80% of the data. The blue dashed line corresponds to the linear model fit between the variance and mean, and the red dashed line represents the fit from a linear model including a quadratic term of the mean in addition to the linear term. The regression coefficient for the quadratic term is highly significant ($P < 2.2e-16$) proving that a Poisson distribution is insufficient to model the overdispersion in the branch lengths. We repeated the linear model fits using the full data observing consistent results (Quadratic term beta $P < 2.2e-16$).

Topology of simulated trees



Fig S3. Topology describing the relationship between branches of simulated trees.
The topology is constructed based on the relationships of 62 mammalian species as reported in the UCSC genome browser.

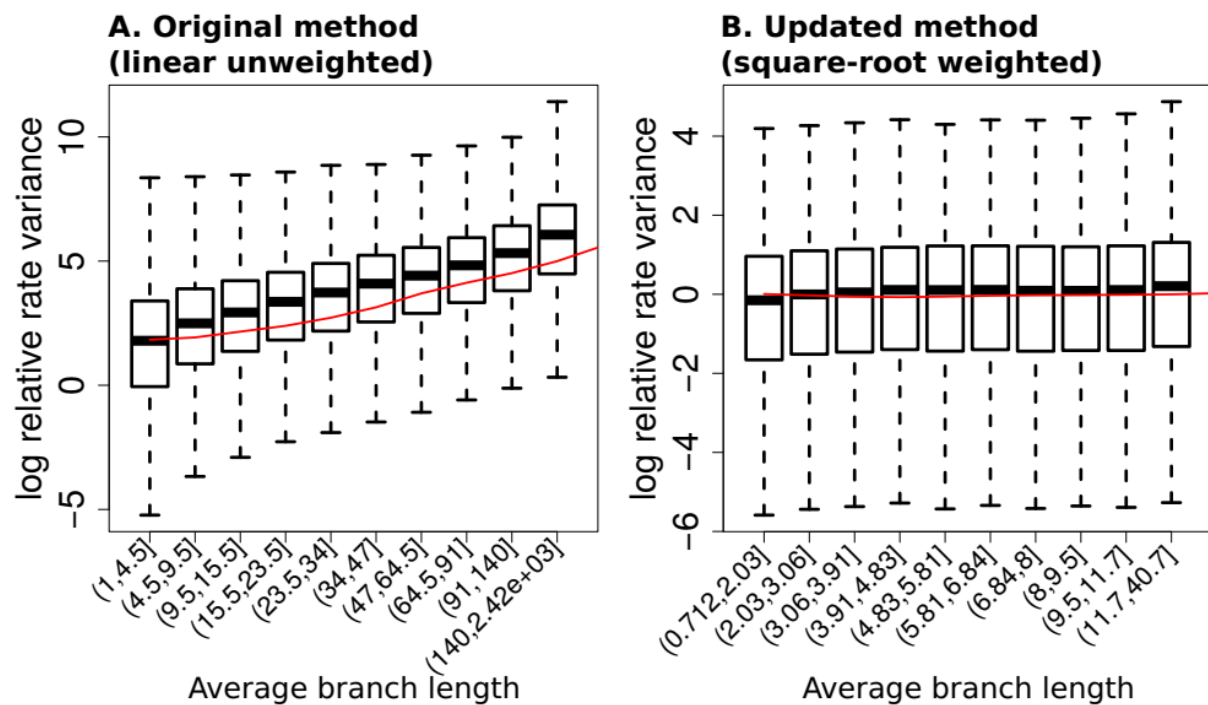


Fig S4. Mean-variance trends in relative rates on branches of simulated phylogenetic trees computed using the two methods. The original method (A) produces heteroscedastic relative rates that show a strong mean-variance trend, whereas relative rates calculated using the updated method (B) show constant variance across branches of different lengths.

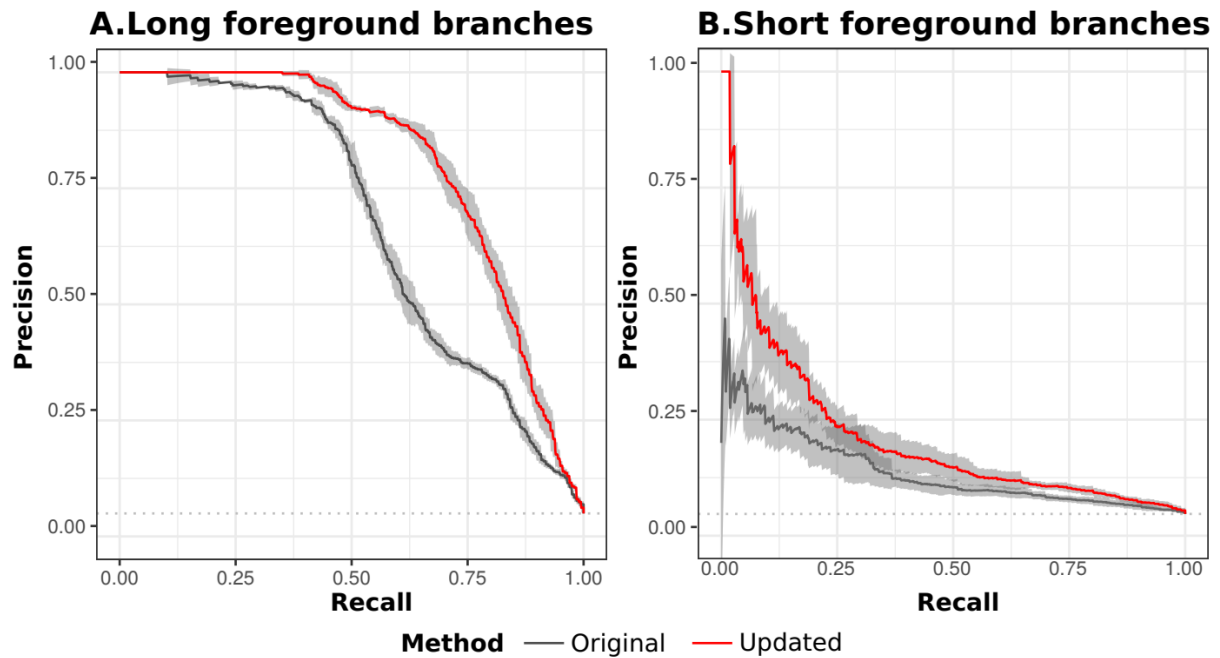


Fig S5. Precision-recall curves for detecting positive trees in simulations. The plots correspond to the precision-recall curves underlying the AUPR distributions in Figure 6B. The curves compare the performance of the updated and original methods to detect positive trees showing convergent rate acceleration. Across five independent simulations of positive and control trees, the updated method outperforms the original method. These results are consistent across varying configurations of foreground branches, including long (A) and short (B) foreground branches respectively. Curves were generated using the R package *precrec* (Saito and Rehmsmeier, 2017). Shaded areas reflect 95% confidence intervals.

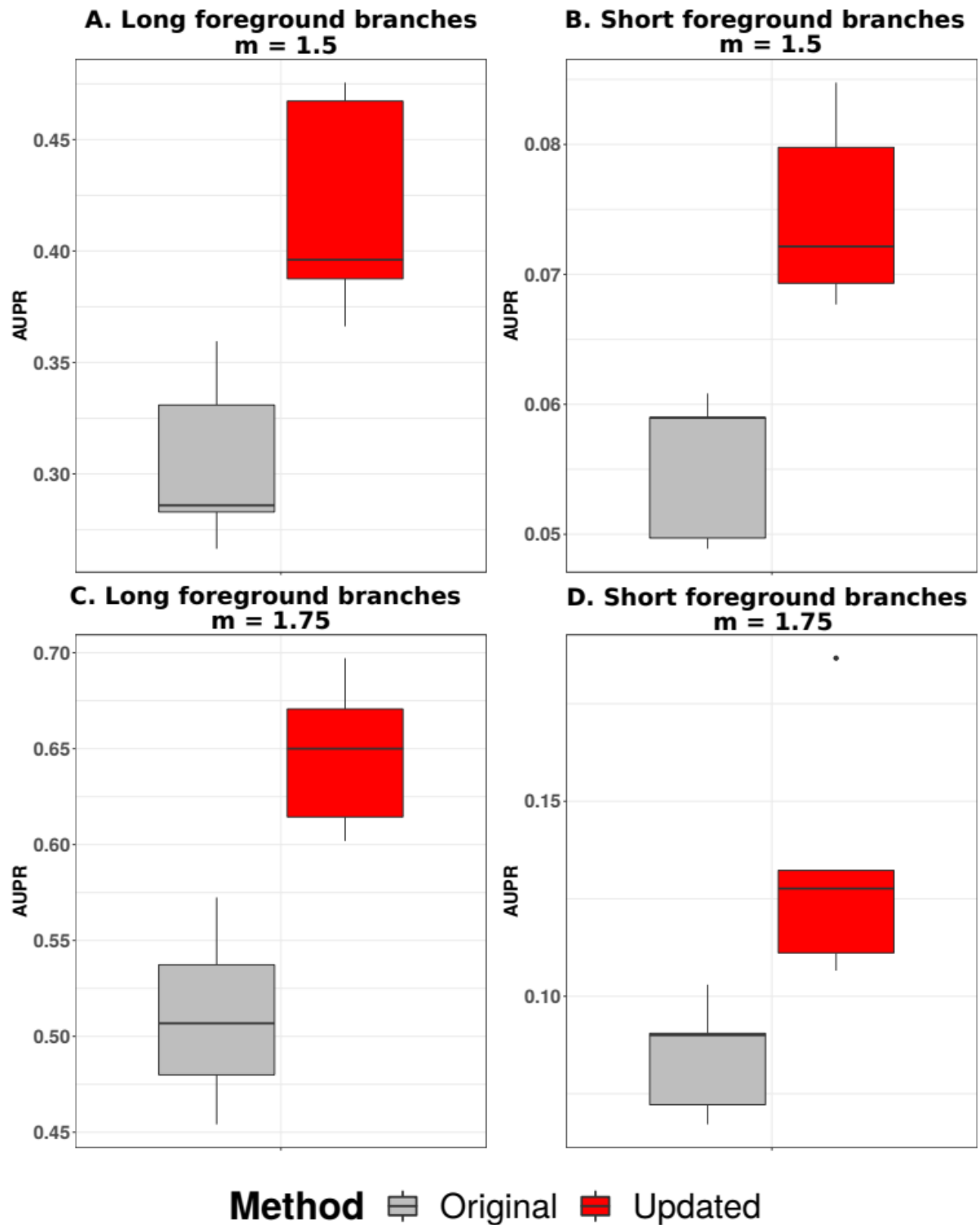


Fig S6. Power to detect rate shifts in foreground branches sampled at varying accelerated rates Across five independent simulations of control and positive trees, we measured the area under precision-recall curve (AUPR) to detect positive trees using the foreground acceleration score. The updated method offers consistently higher power in independent simulations of long (A,C) and short foreground branches (B,D) where the foreground branches are sampled using varying foreground rate multipliers ($m = 1.5$, A and B; $m = 1.75$, C and D).

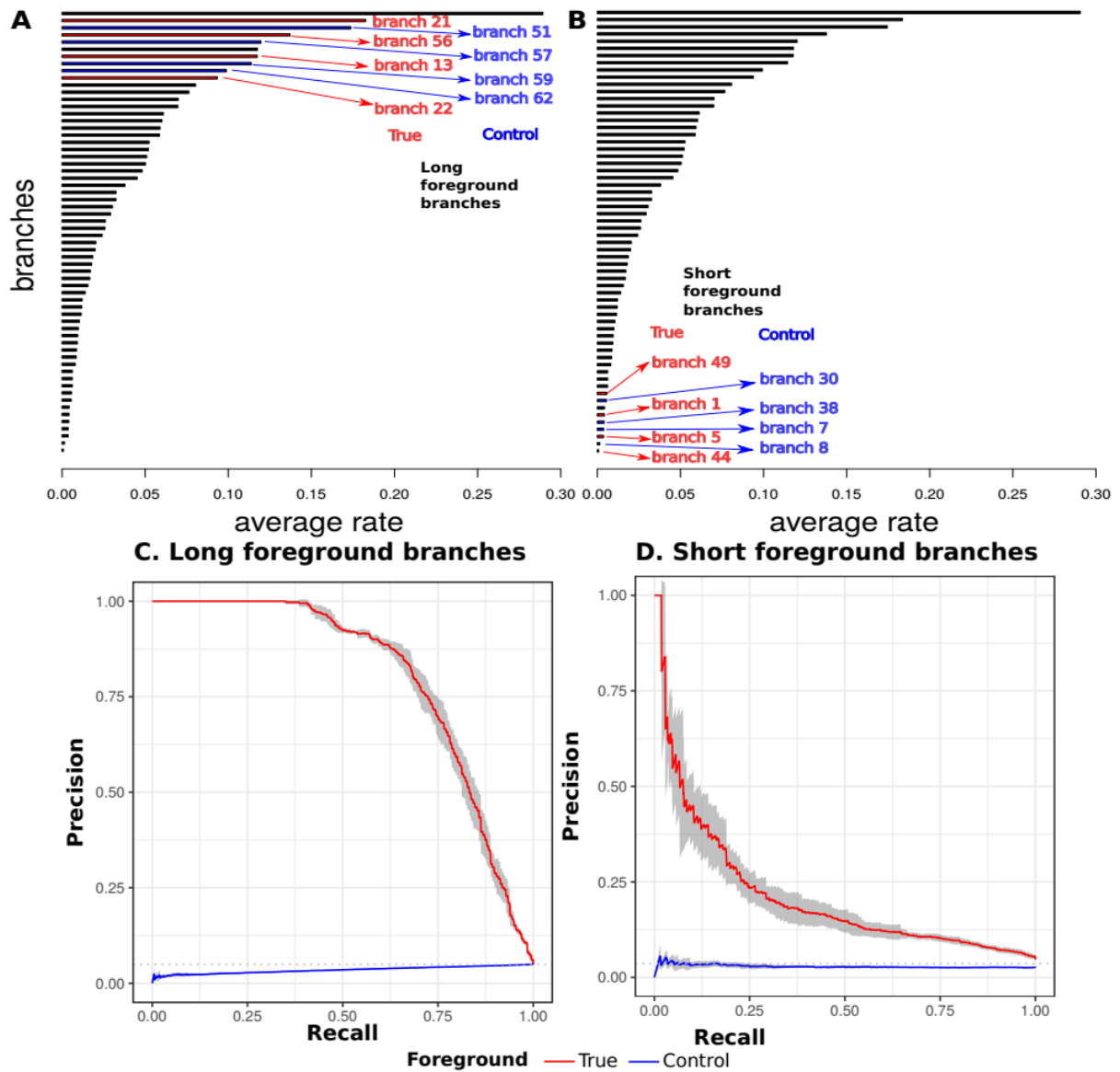


Fig S7. Comparison of power to detect positive trees using true foreground branches vs control foreground branches. Panels A and B correspond to the true and control foreground branches used in independent simulations with long and short foreground branches respectively. The control branches were chosen such that their lengths matched the lengths of the true foreground branches as closely as possible. Panels C and D show the precision-recall curves for detecting positive trees using foreground acceleration scores computed by the updated method. We observed that scores computed using the control foreground branches do not offer any power to detect rate acceleration in the true foreground branches of positive trees. Curves were generated using the R package *precRec* (Saito and Rehmsmeier, 2017). Shaded areas reflect 95% confidence intervals.

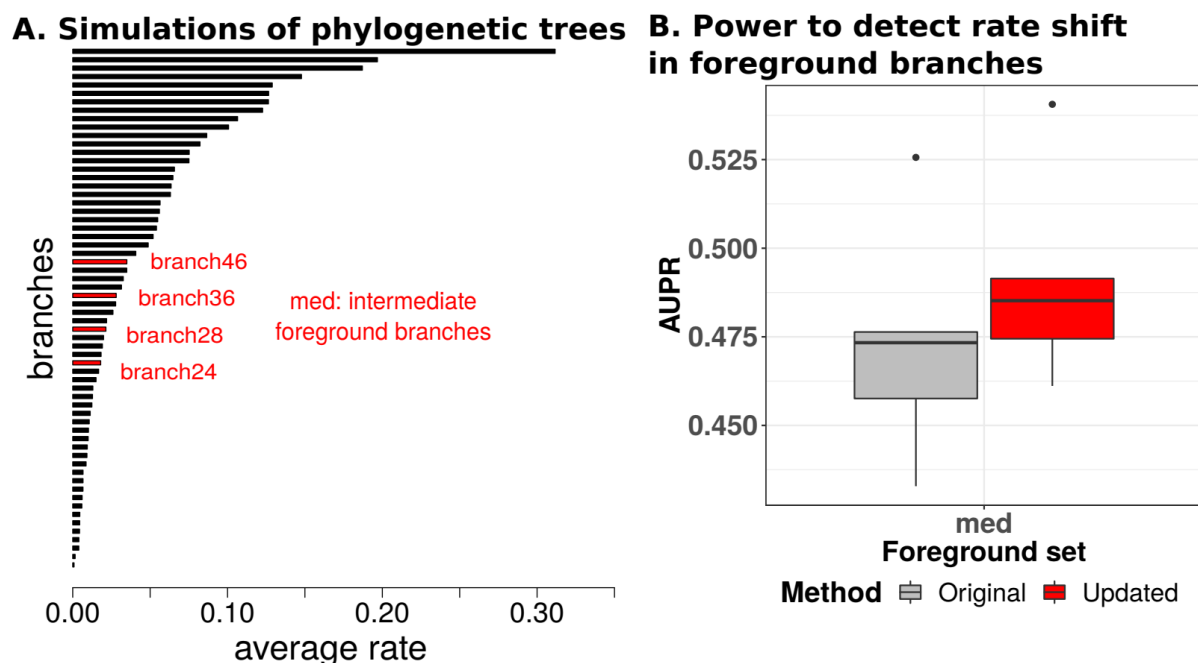


Fig S8. Simulations of phylogenetic trees with foreground branches of intermediate length A. Foreground branches (in red), used for simulating convergent acceleration on intermediately long branches. B. Comparison of power between the two methods to detect convergent rate acceleration on intermediately long foreground branches. Across five independent simulations of control trees and positive trees, we measured the area under the precision-recall curve (AUPR) to precisely detect positive trees using the foreground acceleration score. The AUPR distributions obtained using the updated method to calculate relative rates are significantly elevated compared to the original method.

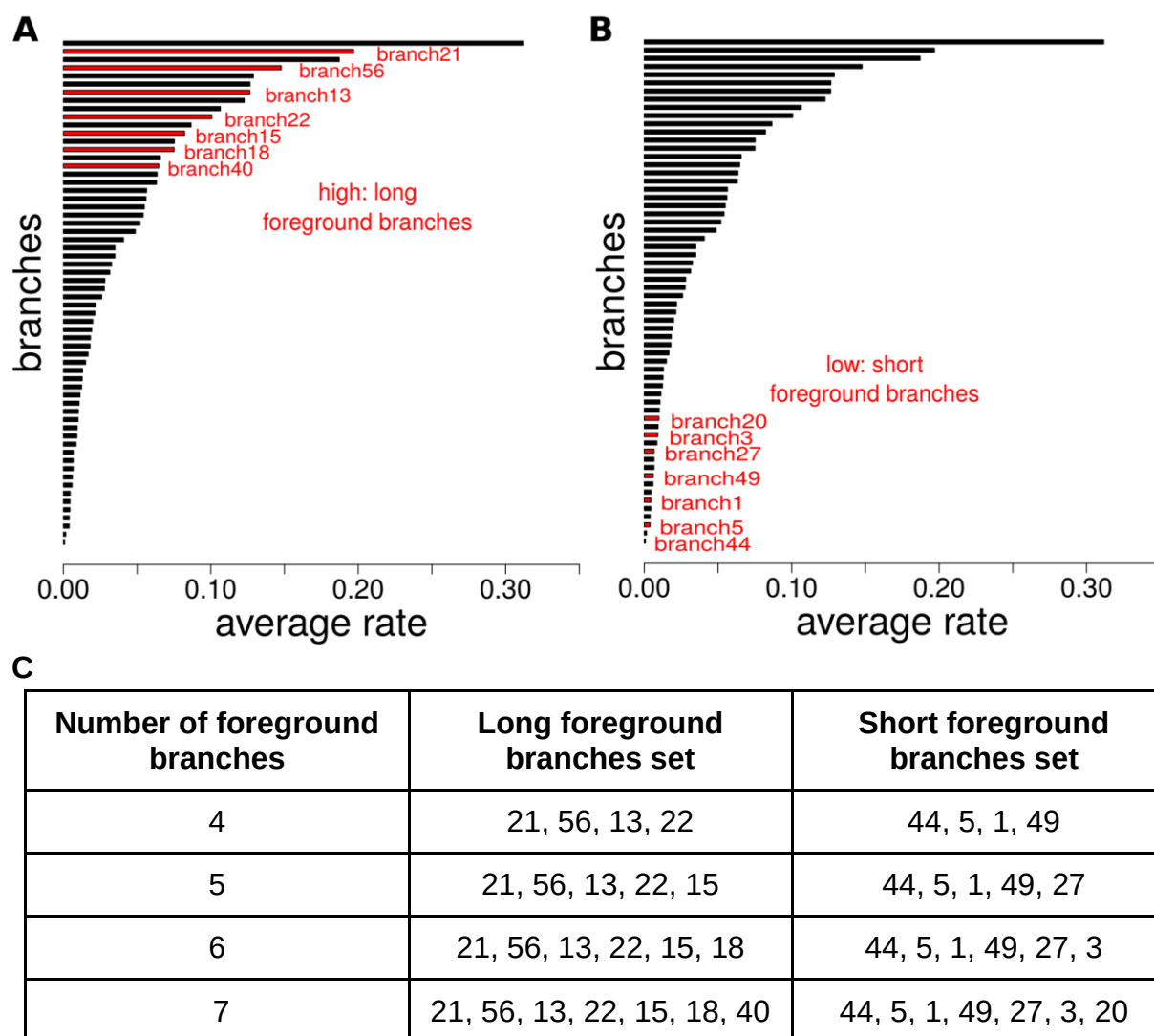
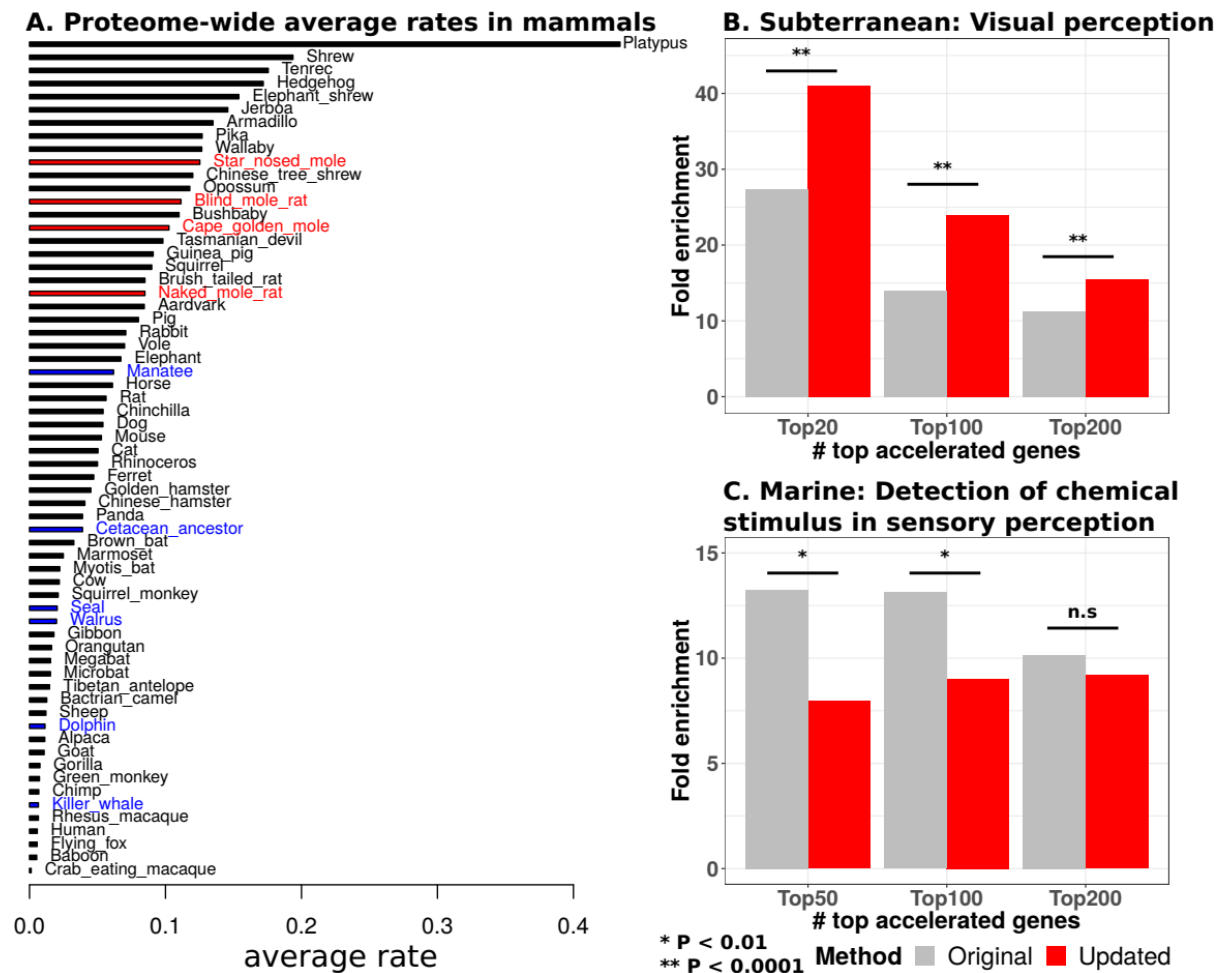


Fig S9. Simulations of phylogenetic trees with varying numbers of foreground branches. Red branches correspond to the seven foreground branches chosen for simulating trees showing convergent rate acceleration on long (A) and short (B) foreground branches. The foreground branch sets of simulated trees used for comparing the power to detect foreground acceleration across different numbers of long and short branches are given in (C).



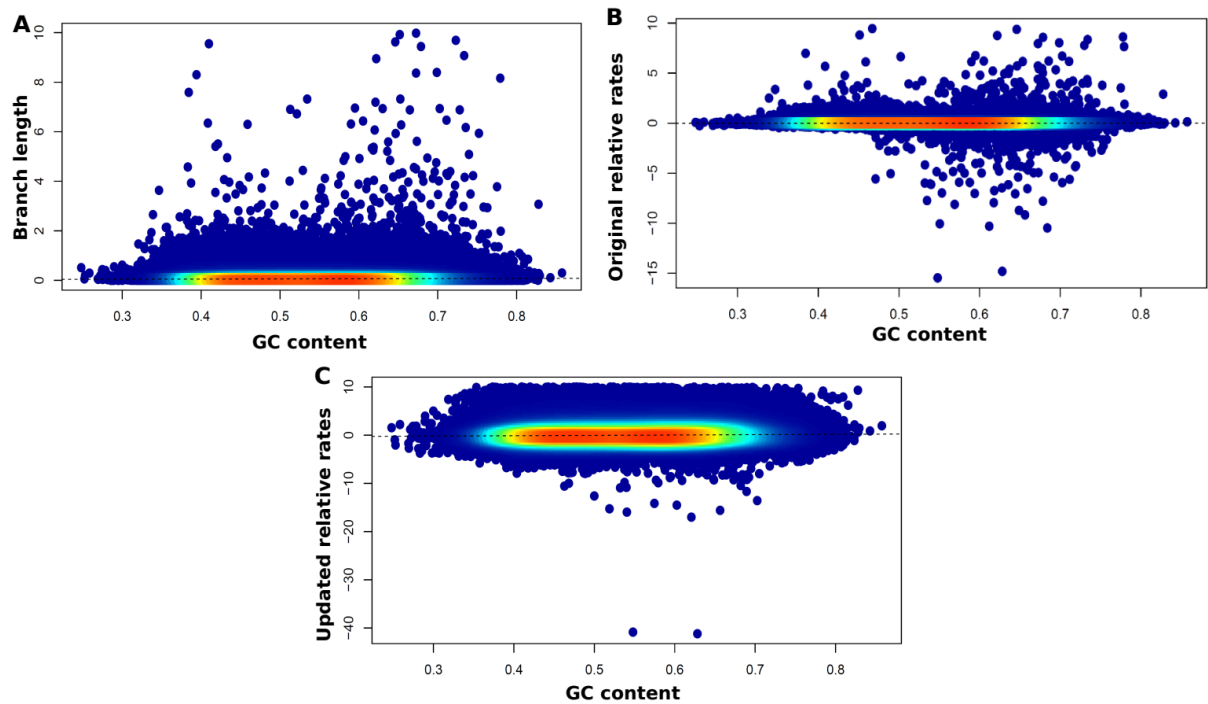


Fig S11: Scatter plot of trends between GC content of sequences and measures of sequence divergence in phylogenetic trees. We measured the GC content (fraction of G or C nucleotides) of extant orthologs from alignments of 19,149 protein-coding genes, and plotted them against the branch lengths (A), relative rates of the corresponding branches computed using original (B), and updated methods (C) respectively. We did not observe any significant global trends in the data with relation to GC content.

References

Saito, T. and Rehmsmeier, M. (2017) 'Precrec: Fast and accurate precision-recall and ROC curve calculations in R', *Bioinformatics*. doi: 10.1093/bioinformatics/btw570.